SS3 to opal: North Pacific blue shark

Bridging the 2022 ISC two-sex assessment to a single-sex Stock Synthesis model, and from it to opal

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1 Overview

The 2022 ISC stock assessment of North Pacific blue shark (*Prionace glauca*) uses Stock Synthesis (SS3 3.30.19) with two sexes. Its base case, S6, models sex-specific growth, natural mortality and selectivity, and fits joint sex-by-length compositions. opal tracks one population vector, so S6 cannot be converted directly.

This case study therefore bridges in two stages:

1. **Within SS3** (Section 3), from S6 to a single-sex SS3 model (B2e) whose structure opal can represent, one change at a time, so the effect of each change is measured on its own.
2. **SS3 to opal** (Section 4), converting B2e with the shared converter in `shared/ss3-to-opal/`, checking the conversion, then fitting opal (Section 5).

Section 6 lists where opal falls short of SS3 or differs from it, and what opal would need to model S6 itself.

In brief:

- S6 re-runs exactly with SS3 3.30.22.1 (NLL 32.2958, SSB_{2020}/SSB_0 0.564).
- The single-sex model B1 fits the data comparably but estimates a much more depleted stock (0.255); switching off the F-ballpark penalty (B2e) moves it back to 0.536. Profiles over $\ln(R_0)$ show why: the stock's scale is weakly determined, and in the single-sex models almost undetermined towards larger stocks.
- The conversion of B2e is exact where opal and SS3 share structure (catch, indices, compositions, biology, unfished equilibrium, selectivity-at-length). opal at SS3′s parameter values differs from SS3 mainly through catch timing: opal removes each year's catch at the start of the year, SS3 continuously.

- Fitted with SS3′s selectivity (O1), opal converges but estimates a much more depleted stock than SS3. Diagnostic refits trace this to how the length compositions are predicted: with SS3′s mid-year length-at-age, or without the compositions, opal's estimate matches SS3′s. Fitting opal's own selectivity (O2) does not converge.

2 The S6 assessment

Table 1: Structure of S6.

Feature	S6 setting
Time	1971-2020, annual, one season (two subseasons), spawning in month 1
Sexes and ages	Two sexes, ages 0-24
Fleets	20 fisheries and 11 surveys
Catch	Weight (mt) except fleets 11, 17 and 18 (1000s of fish)
Biology	Sex-specific age-specific M (fixed); Richards growth with shape 1 (von Bertalanffy) by sex, fixed; length-logistic maturity; 45 pups per mature female
Spawner-recruit	Beverton-Holt, h = 0.613 and sigma_R = 0.4 fixed; main deviations 1971-2020, early deviations 1961-1970; regime offset on the initial equilibrium; initial F on F4
Fishing mortality	Hybrid (method 3)
Selectivity	Size-based double normal (pattern 24) with female offsets from the male curve; F18 cubic spline with a male dogleg; six fisheries and all surveys mirror another fleet; F8 time blocks (1973-81, 1982-92); annual deviations on F19′s peak
Indices	Floating q; extra SD added to the CV; only S5 and S6 (Japanese longline early and late) have lambda 1
Compositions	Joint sex x length (5 cm bins, 5-300 cm), multinomial with a variance adjustment of about 0.002; one size-frequency observation (F11, 1991)
Other likelihood terms	F-ballpark penalty (F = 0.2 in 2013); initial-equilibrium regime

3 The SS3 bridge

`scripts/01-build-ss3-bridge.r` builds each step from the previous step's estimates and re-estimates it with SS3 3.30.22.1 (with the Hessian). Each step makes one change:

- **S6**: as supplied, re-run with 3.30.22.1.
- **B1**: single sex (`Nsexes = -1`); female growth, M and weight-length; compositions summed over sex; no sex-offset selectivity.
- **B2a**: F18′s cubic spline replaced by a double normal.
- **B2b**: F8′s two selectivity blocks become F8 (1973-81) and a new fleet 32 (1982-92).
- **B2c**: no annual deviations on F19′s peak.

- **B2d**: no initial-equilibrium regime offset.
- **B2e**: no F-ballpark penalty.

Table 2: The SS3 bridge. All steps converged (maximum gradient below 2e-4) with a positive-definite Hessian and no parameter on a bound.

Step	Est. par.	NLL	SSB0	SSB2020/SSB0	F2020	SSB MSY
S6	194	32.30	243,875	0.564	0.337	89,706
B1 single sex	127	9.42	274,705	0.255	0.758	80,447
B2a F18 double normal	124	9.60	274,690	0.256	0.757	80,456
B2b F8 split	127	9.54	274,765	0.255	0.758	80,479
B2c no F19 devs	110	10.77	277,354	0.277	0.700	81,354
B2d no regime	109	10.77	277,490	0.276	0.701	81,396
B2e no F ballpark	109	9.16	349,989	0.536	0.345	102,018

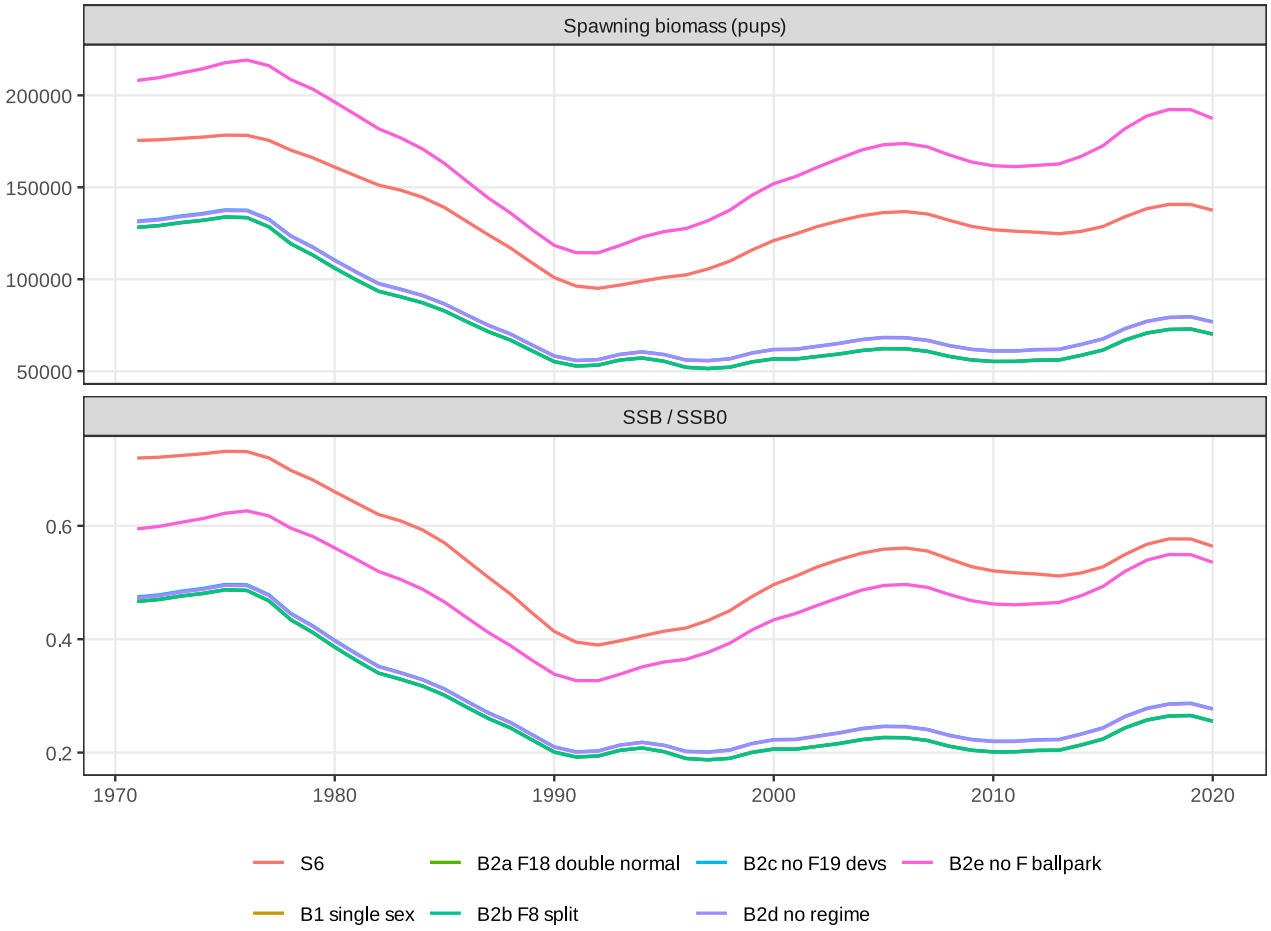


Figure 1: Spawning biomass (top) and depletion (bottom) along the SS3 bridge.

The three groups of steps tell different stories:

- **Single sex (B1)** changes the fit to the compositions most: the length NLL falls from 114.8 to 91.5, at least partly because the combined-sex compositions have half the cells of the sex-by-length ones, so the two NLLs are not directly comparable. Its scale is lower and its terminal depletion 0.255 against S6′s 0.564. The single-sex model uses female growth (245 cm at age 20) for males too (261 cm in S6), and each selectivity curve becomes S6′s male curve.

- **Structural simplifications (B2a-B2d)** change little: the F18 spline, the F8 split and the regime offset each move the likelihood by less than 0.2 units. Removing F19's selectivity deviations costs 1.2 units and moves depletion from 0.255 to 0.277.
- **No F-ballpark (B2e)** moves the likelihood by 1.6 units but the scale a long way: SSB_0 from 277,490 to 349,989 and depletion to 0.536.

3.1 How well is scale determined?

`scripts/02-profile-ss3-r0.r` fixes $\ln(R_0)$ on a grid and re-estimates every other parameter.

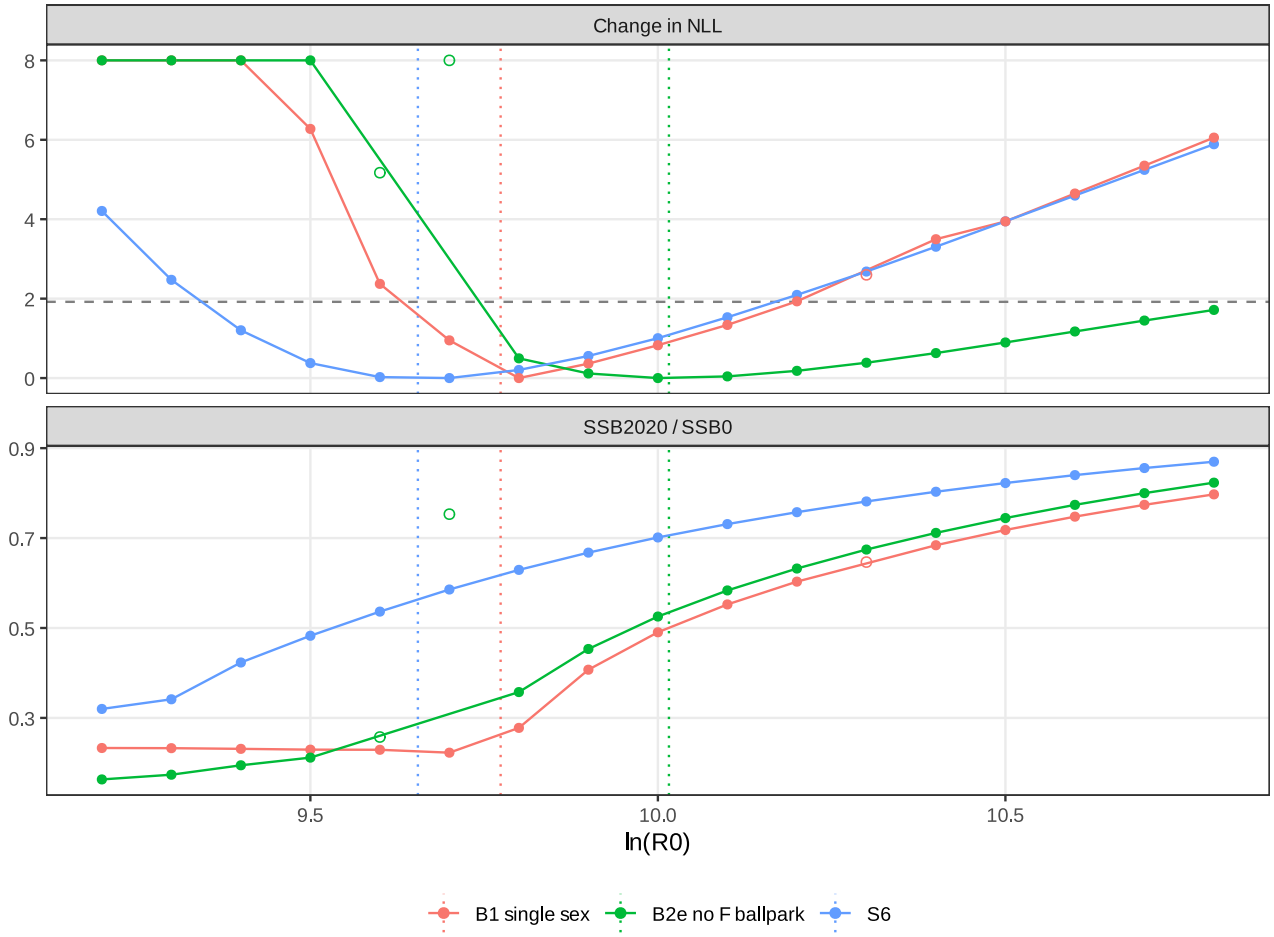


Figure 2: Likelihood profiles over ln(R0) for S6, B1 and B2e (top: change in total NLL, capped at 8, with the 1.92 line of an approximate 95% interval; bottom: terminal depletion). Open points did not converge. The vertical lines are each model's estimate.

Each model's estimate sits at the minimum of its profile, so none is a local optimum. In S6 the approximate 95% interval (change in NLL below 1.92) spans $\ln(R_0)$ 9.4-10.1, depletion 0.42-0.73. In B2e the profile is almost flat towards larger stocks: the interval reaches the edge of the grid ($\ln(R_0)$ = 10.8, depletion 0.82). The index and length components pull in opposite directions and nearly balance. The F-ballpark penalty, worth less than one likelihood unit at the optimum, was what held B1's scale down.

This matters for everything that follows. Small differences between opal and SS3, or between two opal configurations, can move the estimated scale a long way without changing the fit to the data much, so the comparisons below look at the fits to the data as well as at the trajectories.

4 Converting B2e to opal

`config/npo-bsh-b2e.yaml` records the choices the SS3 files do not determine. The opal model has:

- 50 annual time steps, one season, ages 0-24;

- 23 fisheries: SS3′s 21 fishing fleets in SS3 order (F1-F20, then fleet 32, which SS3 numbers after the surveys) and one fishery for each of the two indices kept (S5 and S6; the other nine have lambda 0 in SS3);
- SS3′s age-basis biology (M, weight, maturity, and spawning output as Mat x Fecund x 0.5) from the endgrowth table, at the start of the year. Recruits settle in month 7 (settlement assignment `1 7 1 0`), so at the start of the year SS3′s age-a class is a - 0.5 years past settlement: age 0 is at the 5 cm minimum population size, age 1 at 34.7 cm, and L1 = 64.3 cm (age 1 post-settlement) is reached at mid-year;
- SS3′s parameter values: B_0 (opal's scale parameter, SS3′s SSB_0), h, $sigma_R$, recruitment deviations, the ten early deviations and the 1971 recruitment as the initial age structure, and F4′s initial F.

4.1 Reconciliation

Table 3: Reconciliation with SS3 (exact comparisons). The 21 catch totals are summarised in one row.

Quantity	n	max relative difference	pass
catch totals, 21 fishing fleets	21	0.0e+00	TRUE
CPUE observation count (kept SS3 index fleets)	1	0.0e+00	TRUE
LF total input sample size	1	0.0e+00	TRUE
LF observation rows	1	0.0e+00	TRUE
unfished spawning potential per recruit	1	6.7e-07	TRUE
total weight summed over ages	1	0.0e+00	TRUE

All 38 structural checks pass. The unfished spawning potential per recruit reconciles to 6.7e-07, which confirms in one number the units of M, the female fraction (0.5 for `Nsexes = -1`) and the plus group. opal's unfished numbers-at-age equal SS3′s.

4.2 Selectivity

SS3′s selectivity here is size-based, like opal's, so its double normals carry over exactly rather than being fitted. Where SS3 sets a start or end parameter to −999 ("no rescaling of that limb"), opal's end value is set to the value SS3′s unscaled limb takes at the first or last bin, which makes the two curves identical.

Table 4: Largest difference between opal's starting double normal and SS3's selectivity-at-length, by opal fishery (mirrors and indices take their source fleet's parameters).

opal fishery	SS3 fleet	name	parameters from	max |diff|
1	1	F1_MEX	1	1.9e-06
2	2	F2_CAN	1	1.9e-06
3	3	F3_CHINA	3	6.8e-06
4	4	F4_JPN_KK_SH	4	7.1e-06
5	5	F5_JPN_KK_DP	5	1.7e-06
6	6	F6_JPN_ENY_SHL	4	7.1e-06
7	7	F7_JPN_ENY_DP	7	1.1e-06
8	8	F8_JPN_LG_MESH_EARLY	8	9.0e-06
9	9	F9_JPN_LG_MESH_LATE	9	1.3e-06
10	10	F10_JPN_CST_Oth	9	1.3e-06
11	11	F11_JPN_SM_MESH	11	6.5e-05
12	12	F12_IATTC	1	1.9e-06
13	13	F13_KOREA	3	6.8e-06
14	14	F14_NON_ISC	14	1.1e-05
15	15	F15_USA_GIILL	15	5.9e-06
16	16	F16_USA_SPORT	15	5.9e-06
17	17	F17_USA_Lonline_DP	17	1.0e-05
18	18	F18_USA_Lonline_SH	18	1.6e-05
19	19	F19_TAIW_LG	19	6.0e-06
20	20	F20_TAIW_SM	20	1.4e-05
21	32	F8_JPN_LG_MESH_EARLY_1982	32	2.1e-05
22	25	S5_JPN_EARLY	4	7.1e-06
23	26	S6_JPN_LATE	4	7.1e-06

opal applies the length curves to ages through its own growth (a Schnute curve fitted to SS3's length-at-age; RMSE 3.4 cm, largest error 10.4 cm). SS3 grows fish linearly below age 1 and evaluates fishery selectivity on the mid-year size distribution, so opal's starting selectivity-at-age differs from SS3's even though the curves at length are identical:

Table 5: Largest difference between opal's starting selectivity-at-age and SS3′s, by opal fishery.

fishery	name	max |diff|
1	F1_MEX	0.074
2	F2_CAN	0.074
3	F3_CHINA	0.074
4	F4_JPN_KK_SH	0.082
5	F5_JPN_KK_DP	0.076
6	F6_JPN_ENY_SHL	0.082
7	F7_JPN_ENY_DP	0.078
8	F8_JPN_LG_MESH_EARLY	0.079
9	F9_JPN_LG_MESH_LATE	0.073
10	F10_JPN_CST_Oth	0.073
11	F11_JPN_SM_MESH	0.527
12	F12_IATTC	0.074
13	F13_KOREA	0.074
14	F14_NON_ISC	0.073
15	F15_USA_GIILL	0.074
16	F16_USA_SPORT	0.074
17	F17_USA_Lonline_DP	0.085
18	F18_USA_Lonline_SH	0.065
19	F19_TAIW_LG	0.089
20	F20_TAIW_SM	0.079
21	F8_JPN_LG_MESH_EARLY_1982	0.069
22	S5_JPN_EARLY	0.082
23	S6_JPN_LATE	0.082

4.3 opal at SS3′s values

O0 runs opal at SS3′s parameter values with SS3′s own selectivity-at-age. Nothing is estimated, so any difference from SS3 comes from the two models' dynamics.

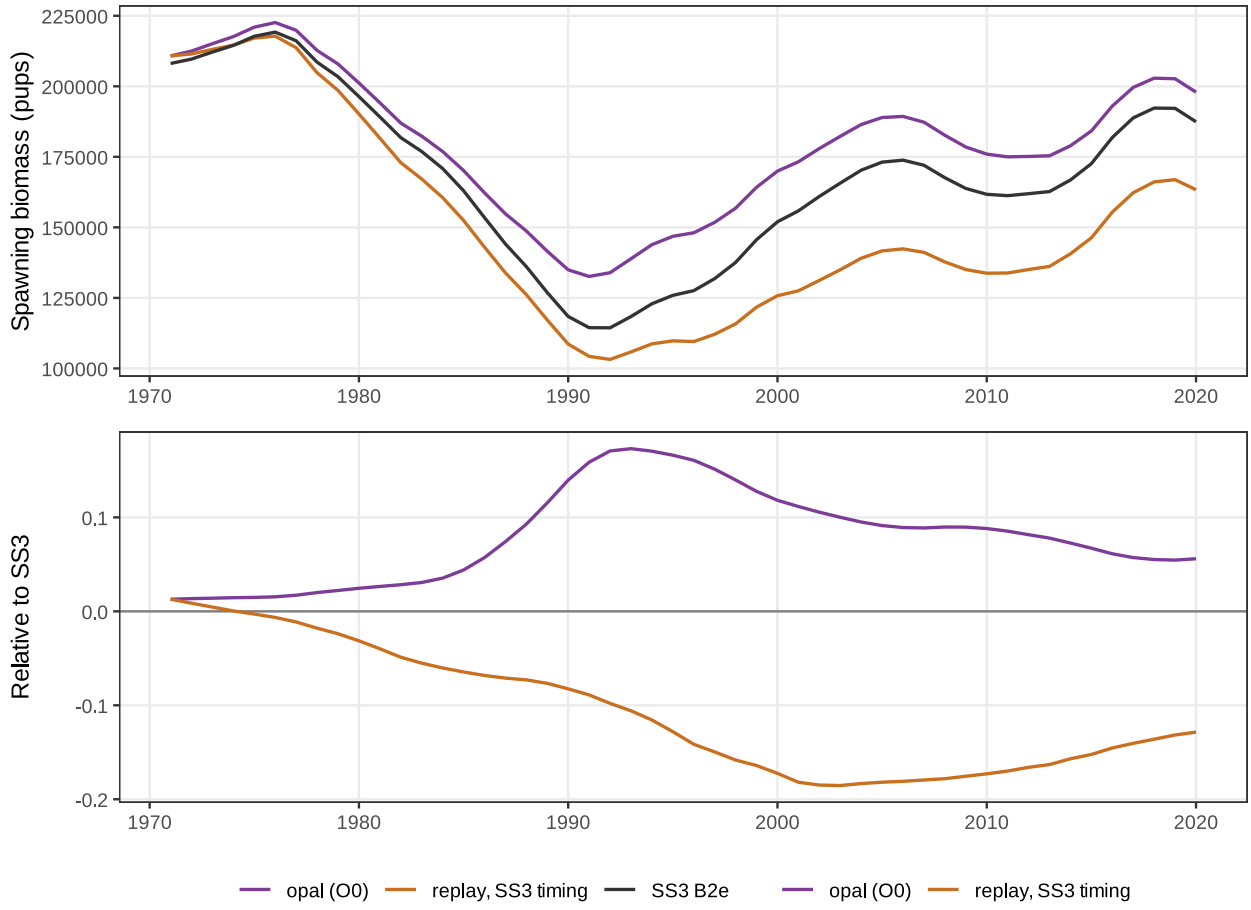


Figure 3: O0: opal at SS3′s values against SS3 B2e (top: spawning biomass; bottom: relative difference). 'Replay, SS3 timing' replays the converted inputs with SS3′s Baranov catch equation.

The first year agrees to 1.3%: the initial equilibrium, initial F and initial deviations are converted directly. Recruitment then tracks SS3 to within a few percent, but spawning biomass drifts up to 17% above SS3 in the high-catch years around 1990-1995. Two timing differences account for this:

- **Catch.** opal removes each year's catch at the start of the year, before natural mortality, as a harvest rate; SS3 removes it continuously (hybrid F). With F_{2020} about 0.34 and M of 0.79 at age 0, the start-of-year removal leaves more fish alive.
- **Catch weight.** SS3 converts catch in weight to numbers with mid-year weights; opal with its single weight-at-age vector (start of year). For the same tonnage opal removes more, smaller fish.

The replay with SS3′s catch equation but opal's start-of-year weights overshoots the other way (19%); with SS3′s mid-year weights it is within 8% of SS3 (tested in `tests/testthat/test-ss3-to-opal-npo-bsh.r`), which places the remaining gap in SS3′s subseason and initial-equilibrium details rather than in the conversion. In opal itself the two effects partly cancel, which is why opal uses start-of-year weights here: mid-year weights would double the gap. `simulations/timing/` studies the catch-timing difference in general.

5 opal fits

Two fits. Growth, h, $sigma_R$, the biology and F4′s initial F stay at SS3′s values; opal's default priors are switched off, as SS3 has no parameter priors. SS3 ties the initial F to an initial-equilibrium catch of 40,000 t (SE 0.05) through a likelihood term opal does not have; left free in opal, the initial F doubled (0.11 to 0.22) and depleted the 1971 stock to 0.37 of B_0.

- **O1** keeps SS3′s selectivity-at-age and estimates B_0, the recruitment deviations, the initial age structure (1971 recruitment and the ten early deviations) and the two catchabilities, starting at SS3′s values.

- **O2** uses opal's own length-based double normals, started at SS3′s curves, and also estimates the selectivity parameters SS3 estimated (peak and widths, within SS3′s bounds on opal's scale); mirrored fleets share their source's parameters, as in SS3. It starts from O1′s estimates.

Table 6: Fits. The objective values are opal's and are not comparable with SS3′s likelihood (different constants, and opal's compositions are predicted differently).

Model	Est. par.	Objective	Max gradient	opal_check()	B0	SSB 2020	SSB2020/B0	Index NLL	Length NLL
SS3 B2e	109				349,989	187,433	0.536		
O0	0	1,547.35			349,989	197,925	0.566	−33.39	1,317.9
O1	64	1,522.97	4.2e-11	pass	272,636	68,937	0.253	−32.53	1,282.6
O2	108	1,428.39	3.1e-01	fail	235,475	33,563	0.143	−27.67	1,173.6
O1-noLF	64	−31.62	5.7e-11	pass	336,369	179,551	0.534	−35.07	1,318.7
O1-mid	64	1,554.99	4.8e-09	pass	319,522	175,548	0.549	−31.51	1,315.8

O1 converged and passes `opal_check()`; its terminal depletion is 0.253 against SS3′s 0.536, although it uses SS3′s selectivity and starts at SS3′s values (O0, depletion 0.566). O2 did not converge (maximum gradient 3.1e-01) and is shown for completeness only: freeing opal's selectivity improves the fit to the compositions by 109 units over O1 while worsening the fit to the indices by 4.9, and the optimiser does not settle (several widths move to the flat ends of the double normal, where the gradient vanishes). Its depletion of 0.143 should not be read as an estimate.

5.1 What moves opal away from SS3

Two diagnostic refits of O1 locate the difference (Table 6, Figure 4):

- **Without the length compositions** (O1-noLF, lambda 0), opal's estimate matches SS3′s: terminal depletion 0.534, B_0 336,369. opal's dynamics, the indices and the recruitment penalty agree with SS3′s; the compositions are what pull O1 down.
- **With SS3′s mid-year length-at-age** in place of opal's growth when the compositions are predicted (O1-mid), depletion is 0.549, close to SS3′s. SS3 predicts a fishery's length composition from its catch over the year; opal predicts it from numbers at the start of the year with start-of-year lengths. Because recruits settle in month 7, SS3′s age-1 class is only half a year past settlement at the start of the year (34.7 cm) and reaches L1 (64.3 cm, age 1 post-settlement) at mid-year; the gap is 13 cm at age 2 and shrinks with age. With SS3′s start-of-year length-at-age instead the fit runs away (B_0 over 10^7), and opal's own Schnute growth (fitted to the start-of-year lengths, linear growth below age 1 aside) lands in between.

O1-mid uses `ss3_patch_opal_external_laa()`, a local edit of opal's model function that substitutes a supplied length-at-age; it is a diagnostic, not a way to fit opal. What opal would need is length-at-age, and the age composition of the catch, at the time each observation is made (Section 6).



Figure 4: Spawning biomass, depletion and age-0 recruitment: SS3 B2e and the opal models.

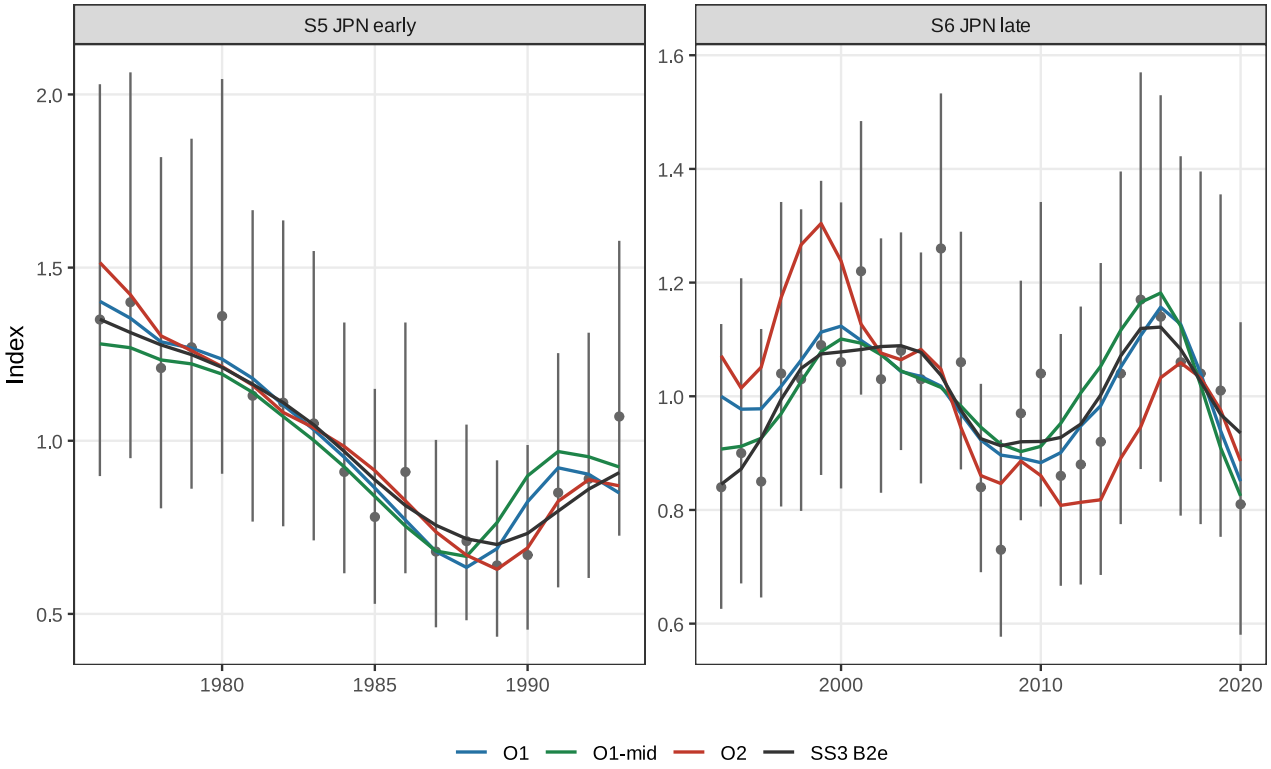


Figure 5: Fits to the two indices (points: observed, with 95% intervals from the input SE).

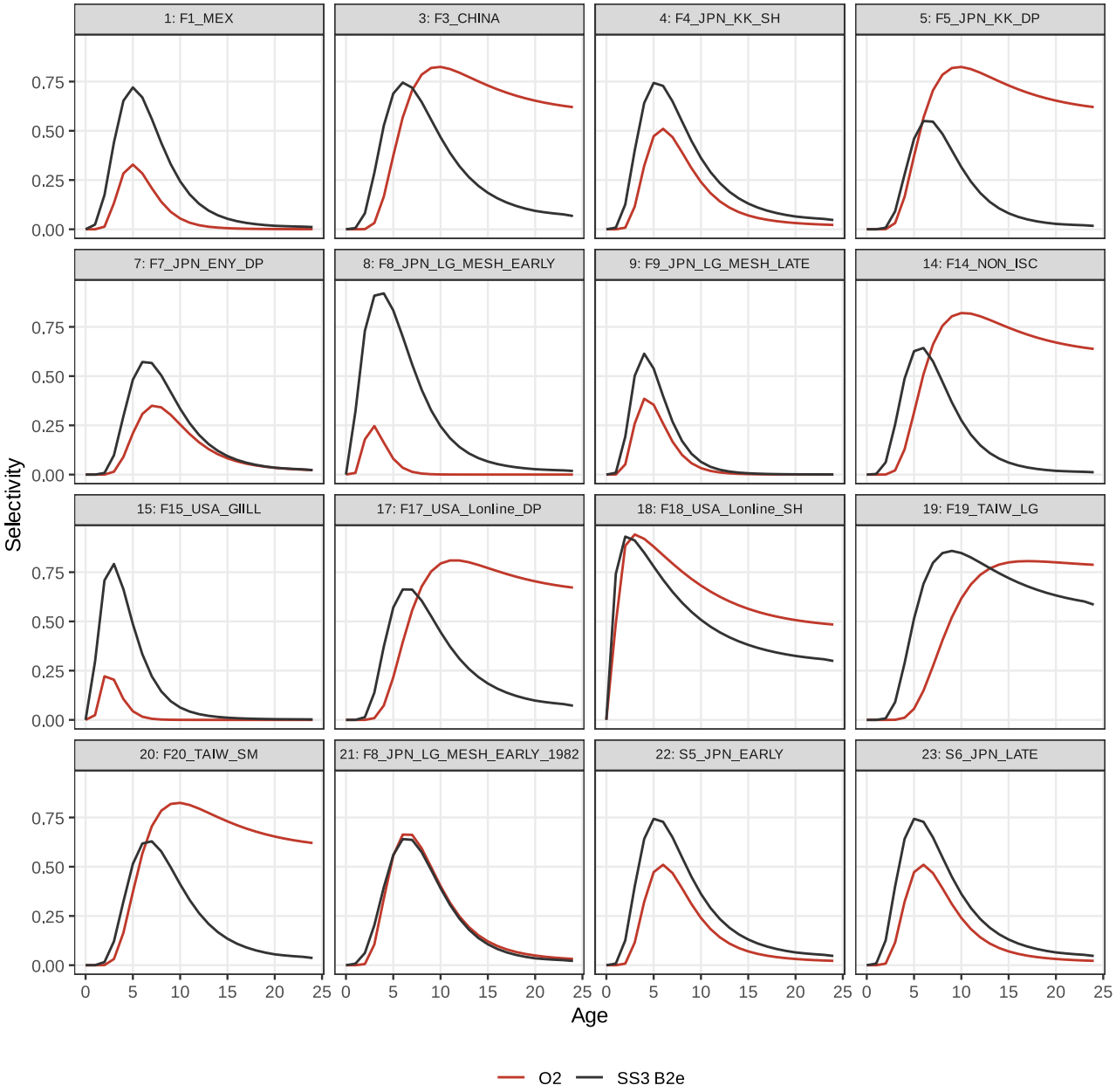


Figure 6: Selectivity-at-age in 2020: SS3 B2e (black) and opal O2 (red; not converged), for the opal fisheries with length data and the two indices.

6 Limitations and differences from SS3

6.1 Not available in opal

- **Two sexes.** opal tracks one population vector and scales spawning potential by a sex ratio. Modelling S6 needs a sex dimension in the numbers at age, sex-specific growth, M and selectivity (including SS3′s offset parameterisations), sex-specific or joint sex-by-length compositions, and spawning biomass from females only.
- **Cubic-spline selectivity** (S6′s F18). opal has logistic, double normal and double Richards curves, or a fixed external matrix.
- **Time-varying selectivity.** Blocks (F8) and deviations (F19) cannot be estimated. Blocks that replace parameters are reproduced exactly by splitting the fleet (B2b); deviations have no equivalent (B2c).
- **Initial-equilibrium catch.** SS3 fits the catch implied by the initial F (40,000 t for F4 here) as data; opal has no such term, so its initial F is fixed at SS3′s value.

- **Regime offset on the initial equilibrium recruitment**, and SS3′s **F-ballpark** penalty. opal fits catch exactly with harvest rates, so it does not need the latter for convergence, but as Section 3 shows the penalty carried information on scale.
- **Generalised size-frequency data** on irregular bins as a separate data type with its own weighting (one F11 observation here, not carried).

6.2 Structural differences handled in the conversion

- **Catch timing.** opal removes each season's catch at its start; SS3 here removes it continuously (hybrid F). This is the largest difference at SS3′s values (Figure 3). Sub-annual opal seasons with the catch split across them would approximate continuous removal, but opal's single weight-at-age vector would still apply start-of-year weights in every season.
- **Observation timing.** SS3 observes the indices in month 7 and the fishery compositions over the year (catch-weighted, with mid-year lengths); opal predicts both from numbers and lengths at the start of the season. For this annual model, with recruits settling in month 7 and start-of-year lengths 30 cm below mid-year lengths at age 1, this is the difference that matters most once opal is fitted: it alone moves O1′s terminal depletion from about 0.55 to 0.25 (Section 5). opal would need observation-time length-at-age (growth evaluated at the fraction of the year or season when the fleet fishes) and catch-weighted predicted compositions; sub-annual seasons would approximate both.
- **Growth below the first reference age and settlement timing.** SS3 grows fish linearly from settlement (month 7, at the 5 cm minimum size) to age 1 post-settlement; opal has no settlement time and its Schnute curve, fitted to SS3′s start-of-year lengths, extrapolates von Bertalanffy growth below age 1. On the endgrowth biology route this only affects the PLA used for selectivity.
- **Extra index SD.** SS3 adds it to the input SE; opal combines its own tau in quadrature. The converter adds SS3′s value to the input SE and opal's tau stays at zero.
- **Catchability.** SS3 solves it analytically (floating q); opal estimates it.
- **Fleet layout.** SS3 3.30 allows fishing fleets after surveys (fleet 32 here); the converter maps SS3 fleets to opal fisheries by fleet type.

6.3 What the bridge itself changes

The single-sex step is not neutral: B1′s stock is smaller and more depleted than S6′s. Female growth is applied to males (who are larger), each selectivity curve becomes S6′s male curve, and the compositions lose their sex information. A sex-averaged biology (the planned sensitivity B1b) would show how much of the change comes from growth and M rather than from the selectivity and the compositions.

7 Reproducing this document

Step	Script	Writes
1	`scripts/01-build-ss3-bridge.r`	the SS3 bridge models and `vignettes/data/npo-bsh-ss3-bridge-cache.rds`
2	`scripts/02-profile-ss3-r0.r`	`vignettes/data/npo-bsh-ss3-r0-profile-cache.rds`
3	`scripts/03-convert-fit-npo-bsh-opal.r`	`vignettes/data/npo-bsh-opal-cache.rds` and the fitted models
4	`quarto render vignettes/npo-bsh-ss3-to-opal.qmd`	this document

```
Caches: bridge 2026-10-28 03:24:48 UTC (SS3 3.30.22.1); opal 2026-10-28 04:46:45 UTC (opal
0.1.1, r4ss 1.56.0, R 4.5.3).
```